



Republic of Rwanda
Ministry of Education



SCAFFOLDING OPERATIONS

BDCS0501

Erect Scaffolding

Competence

RQF Level: 5

Learning Hours



Credits: 7

Sector: Construction and Building Services

Trade: Building Construction

Module Type: Specific

Curriculum: CBSBDC5001-TVET CERTIFICATE V IN BUILDING CONSTRUCTION

Copyright: © Rwanda TVET Board, 2024

Issue Date: May 2024

2024-25

Purpose statement	This specific module describes the skills, knowledge, and attitudes required to erect scaffolding. It is meant for learners who have successfully completed level IV in building construction and pursuing level V in building construction. At the end of this module, learners will be able to perform preliminary works, assemble the scaffolding units, Monitor use of scaffolding and dismantle the scaffolding units.					
Learning assumed to be in place	N/A					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		25%	Formative assessment	30%	50%
	Practical work:		75%		70%	
	• Group project and presentation	25%				
	• Individual project /Work	50%				
			Summative Assessment		50%	

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Perform preliminary works	1.1 Scaffolding is adequately Identified based to the site conditions
	1.2 Materials, tools, and equipment are effectively selected according to the specifications
	1.3 Working place is adequately prepared based on the site condition
2. Assemble the scaffolding units	2.1 A frame of scaffold is systematically assembled in accordance with the required techniques
	2.2 Planks are systematically placed according to the design
	2.3 Guard rails are appropriately installed in accordance with safety measures
	2.4 Access to the scaffolding is correctly prepared according to working environment
3. Monitor use of scaffolding	3.1 Safety measures are effectively applied according to the rules and regulations
	3.2 Scaffolding is regularly checked according to the types of scaffolds
	3.3 Scaffolding is properly maintained based on monitoring reports.

4. Dismantle the scaffolding units	4.1 Scaffold components are systematically disassembled according to its type
	4.2 Scaffold components are systematically arranged as per members
	4.3 Scaffolding elements are appropriately stored in according to their types

Intended Knowledge, Skills and Attitude

Knowledge	Skills	Attitude
✓ Describe the types of construction drawings used in scaffolding works ✓ Identify materials, tools and equipment ✓ Identify methods of scaffolding setting out ✓ Describe the material's properties and applications ✓ Describe knowledge required in requesting of construction works resources ✓ Describe knowledge required to ensure environmental sustainability in scaffolding erection ✓ Explain timber, wood and steel technology ✓ Define quality control and assurance principles for scaffolding erection ✓ Identify safety and health regulations for scaffolding works	✓ Apply safety precautions in scaffolding works ✓ Interpret design drawings, convention symbols, and signage. ✓ Analyze the site requirements ✓ Identify potential hazards and risks. ✓ Identify, locate and report environmental issues ✓ Use of scaffolding equipment ✓ Assemble scaffolding frame ✓ Fix members/accessories of scaffolding ✓ Secure the security of the scaffolding users ✓ Perform technical drawing interpretation ✓ Apply measurement skills ✓ Using hand tools to take measurements ✓ Perform data collection, recording, and calculation	✓ Reasonable assumption ✓ Use materials efficiently and effectively ✓ Use creativity and innovation throughout the design works ✓ Demonstrate punctuality during the implementation of design project ✓ Demonstrate resourcefulness in the new design trends ✓ Be attentive to detail in the calculation of quantities, estimation of costs, and assessment of project specifications ✓ Demonstrate effective communication skills to stakeholders. ✓ Manage time effectively to meet project timelines. ✓ Having teamwork and collaboration spirit ✓ Demonstrate ethical practices at the workplace ✓ Demonstrate willingness to adapt to changes in scaffolding works

✓ Identify techniques for assembling scaffolding ✓ Identify specifications of resources ✓ Identify measurement tools and equipment ✓ Identify cleaning methods	✓ Evaluate the valuation of the building for different specifications. ✓ Estimate cost of construction work ✓ Monitor the project cost ✓ Apply reporting skills ✓ comply with laws, regulations and standards ✓ Manage scaffolding resources ✓ Protect environment and ground conditions	✓ Practice compliance with health, safety, social and environmental regulations ✓ Listen to clients' needs open-mindedly ✓ Pay attention to details while erecting scaffolding frame
---	--	--

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Perform preliminary works 2. Assemble the scaffolding units 3. Monitor use of scaffolding 4. Dismantle the scaffolding units
--------------------------	---

Learning outcome 1: Perform preliminary works	Learning hours: 10 hours
Indicative content	
• Identification of scaffoldings ✓ Types of scaffolding based on support ✚ Dependent scaffoldings ✚ Independent scaffoldings ✓ Identify types of scaffolding based on materials	

 Steels  Aluminium  Timber ✓ Conduct site visit  Inspect site conditions  Review site and design plan • Selection of materials, tools and equipment use to erect scaffolding. ✓ Classification of Materials ✓ Classification of Tools ✓ Classification of Equipment • Preparation of a Working place for scaffolding ✓ Clear and clean workplace ✓ Protect workplace	
Resources required for the indicative content	
Equipment	• Lifting machine, Wheelbarrow, Truck, Crane truck, PPE's
Materials	• Oil, Tubes, couplers, Boards, Ledgers, ladders, standards, scaffolding, diagonal braces, braces, working platform, transoms, putlogs, soleplate, platform
Tools	• Spanner, Ratchet, Spirit level, Wrench, Measuring tape, Nipper -cutter, Scaffolding tag box, Cable ties, Plastic cover for coupler bolts, Plastic caps for scaffold tube, tool box, hammer
Facilitation techniques	Brainstorming, Demonstration and Simulation, Practical exercise, Field study. • The trainer organizes small groups in the classroom and he/she asks students to discuss types of scaffolding and present their findings • On field, trainer forms groups of 4-6 trainees and asks trainees to perform the following tasks: ✓ Conduct site visit for workplace preparation ✓ Prepare resources ✓ Prepare workplace for scaffolding (Metal, Aluminium and timber) • The teacher allows time to each group to independently repeat the task while trainer monitors activity and provides support where needed • The teacher wraps up with questions, answers and feedback to summarize and conclude the session.
Formative assessment methods	• Written based assessment • Oral presentation • Product assessment

Learning outcome 2: Assemble the scaffolding units		Learning hours:20 hours
Indicative content		
• Assemble of scaffold frame ✓ Identification of scaffolding parts ✚ Standards ✚ Ledgers ✚ Braces ✚ Putlogs ✚ Boarding or platforms ✚ Guardrails ✚ Toe boards ✚ Wheels (automated and non-automated) ✓ Assembling scaffolding methods ✚ Nailing methods ✚ Threading methods ✚ Automated assembly method (Robotic and sensors) • Placing Planks on the frame ✓ Classification of scaffold planks ✚ By Material (Metal planks & Wooden planks) ✚ By Design (Single & Double) ✓ Preparing planks for scaffolding. ✚ Measuring ✚ Marking ✚ Laying ✚ Fixing • Installation of Guard rails and access to the scaffolding ✓ Performing guard rail installation ✚ Measuring ✚ Marking ✚ Fixing ✚ Inspection of fixed guard rails ✓ Providing access to the scaffolding ✚ By ladders ✚ By stairs ✚ By ramps or direct passage from another scaffold		
Resources required for the indicative content		
Equipment	• Lifting machine, Wheelbarrow, Truck, Crane truck, PPE's	
Materials	• Oil, Tubes, couplers, Boards, timber, metals,	

Tools	• Spanner, Ratchet, Spirit level, Wrench, Measuring tape, Nipper -cutter, Scaffolding tag box, Cable ties, Plastic cover for coupler bolts, Plastic caps for scaffold tube.
Facilitation techniques	Brainstorming, Demonstration and Simulation, Practical exercise, Field study. • In classroom with manageable groups, trainees discuss about scaffolding assembling procedures, they. • On field, the trainer forms groups of 3-5 trainees and asks trainees to perform the following tasks: ✓ Interpret drawings and identify required resources ✓ Prepare resources ✓ Assemble scaffold frame (Metal, Aluminium and timber) • The teacher allows time to each group to independently repeat the task while trainer monitors activity and provides support where needed. • The teacher wraps up with questions, answers and feedback to summarize and conclude the session.
Formative assessment methods	• Written based assessment • Oral presentation • Product assessment • Performance assessment

Learning outcome 3: Monitor use of scaffolding	Learning hours: 20 hours
Indicative content	
• Applying safety measures on scaffolding. ✓ Scaffolding Safety guidelines ✓ Identification of major hazards associated with scaffolding ✚ Fall hazard ✚ Electrocution hazards ✚ Falling object hazards ✚ Caught-between hazards ✚ Collapse Hazards ✚ Weather hazards ✚ Lighting Hazards ✚ Equipment Hazards ✓ Prevent scaffolding hazard • Checking the stability of Scaffolding ✓ Check the assembling of scaffolding ✓ Check damage to the scaffolding ✓ Check suitability and functionality • Maintenance of scaffolding.	

✓ Clean scaffolding ✓ Prevent Rust/corrosion and Deterioration ✓ Keep wood and moving parts covered. ✓ Replace any Faulty or damaged Parts	
Resources required for the indicative content	
Equipment	• Lifting machine, Wheelbarrow, Truck, Crane truck, PPE's
Materials	• Oil, Tubes, couplers, Boards
Tools	• Spanner, Ratchet, Spirit level, Wrench, Measuring tape, Nipper -cutter, Scaffolding tag box, Cable ties, Plastic cover for coupler bolts, Plastic caps for scaffold tube.
Facilitation techniques	Brainstorming, Demonstration and Simulation, Practical exercise, Field study. • In classroom with manageable groups, trainees discuss about scaffolding monitoring and their construction procedures. • On field, trainer forms groups of 3-5 trainees and asks trainees to perform the following tasks: ✓ Interpret drawings and identify required resources ✓ Prepare resources accordingly ✓ Maintain scaffold (Metal, Aluminium and timber) • The trainer allows time to each group to independently repeat the task while trainer monitors activity and provides support where needed. • The trainer wraps up with questions, answers and feedback to summarize and conclude the session.
Formative assessment methods	• Written based assessment • Oral presentation • Product assessment • Performance assessment

Learning outcome 4: Dismantle the scaffolding units	Learning hours:20 hours
Indicative content	
• Disassembling scaffolding components ✓ Assessing the scaffolding component installation ✓ Dismantle scaffolding components ✓ Arrangement of Scaffold components ✓ Applying automated disassembly method (Robotic and sensors) • Storing of Scaffolding elements. ✓ Identifying factors to consider before storing scaffolding	

- ✓ Assessment of condition
- ✓ Documentation and inventory
- ✓ Compliance with regulations
- ✓ Consideration of weight and dimensions
- ✓ Accessibility

- ✓ Applying storing procedures for scaffolding components
- ✓ Applying Methods of cleaning the Working area

-  Manual
-  Mechanical

Resources required for the indicative content

Equipment	PPE's, Hoists and Pulleys, ropes, wheelbarrow
Materials	Oil, Steel/aluminum scaffoldings, ladders
Tools	Adjustable Wrenches, Box end wrenches, spanners, hammers, mallets, tool box
Facilitation techniques	Brainstorming, Demonstration and Simulation, Practical exercise, Field study. • In classroom with manageable groups, trainees discuss about scaffolding dismantling and the storing of materials. • On field, trainer forms groups of 4-6 trainees and asks trainees to perform the following tasks: ✓ Interpret drawings and identify required resources ✓ Prepare resources ✓ Dismantling of scaffolding (Metal, Aluminium and timber) ✓ Finishing process • The teacher allows time to each group to independently repeat the task while trainer monitors activity and provides support and guidance where needed. • The teacher wraps up with questions, answers and feedback to summarize and conclude the session.
Formative assessment methods	• Written based assessment • Oral presentation • Product assessment • Performance assessment

Integrated/Summative assessment (For specific module)

Integrated situation

XYZ hospital located in Rusizi district has a challenge of old building and wishes to perform routine maintenance and inspection on its old buildings, the old building is a G+1. This hospital would like to hire a scaffolder to erect scaffolding around the building to provide safe access for workers to access all levels of the building. The building has a height of 6.30 m and the length of the scaffolding is 2m.

1. Determine the height of the scaffolding to use
2. Assemble the scaffolding for workers to access the building.
3. Disassemble the scaffolding after completion of the work.

Resources

Tools	Spanner, Ratchet, Spirit level, Wrench, Measuring tape, Nipper -cutter, Scaffolding tag box, Cable ties, Plastic cover for coupler bolts, Plastic caps for scaffold tube.
Equipment	Lifting machine, Wheelbarrow, Truck, Crane truck, PPE's
Materials/ Consumables	Oil, Steel/aluminum scaffoldings components, ladders

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Perform preliminary works	Scaffolding is adequately identified based on on-site conditions	The type of scaffolding is identified			4
		site conditions are inspected			4
	Materials, tools and equipment are effectively selected according to the specifications	Materials are selected			4
		Tools are selected			4
		Equipment are selected			4
	Working place is adequately prepared based on the site condition	Workplace is prepared			4
		Hazards are removed			5
Assemble the scaffolding units	A frame of scaffold is systematically assembled in accordance with the required techniques	scaffolding parts are identified			7
		Scaffolding parts are fixed			7
		The stability of scaffolding is checked			7
	Planks are systematically placed according to the design	Planks are placed on the frame of the scaffolding			7
	Guard rails are appropriately installed/ in accordance with safety measures	Guardrails of scaffolding is installed			7

	Access to the scaffolding is correctly constructed/ according to working environment	Access to the scaffolding are provided			6
Monitor use of scaffolding	safety measures are effectively applied according to the rules and regulations	Rules and regulations of using scaffolding are respected			6
Dismantle the scaffolding units	Scaffold components are systematically disassembled according to its type	Scaffolding components are disassembled			6
	Scaffold components are systematically arranged as per members	Scaffolding components are Arranged			6
		Scaffolding components are stored			6
		Working area is properly cleaned according to the type of wastage	Workplace area is cleaned		
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

References

1. www.proconstructionguide.com/how-to-install-frame-scaffolding/
2. The Importance of Guardrail Systems in Scaffolding Safety (sunbeltrentals.com)
3. access to scaffolding - Search (bing.com)
4. Scaffolding Safety: Tips & Tagging | Safety Culture
5. <https://ausf.com.au/guide-for-scaffolding-maintenance-and-repair/>
6. <https://aaitsccaffold.com/blog/essential-scaffolding-maintenance-tips-for-a-safer-workplace/>
7. <https://www.turboscaffolding.com.au/blog/a-comprehensive-checklist-to-maintain-scaffolding-and-scaffold-supplies/>
8. Types of Scaffolding Planks - Kennison Forest Products, Inc.
9. BS 1139-1 2:1990 Metal scaffolding Tubes-Specification of aluminum tube